BIOT 6900 Module 6, Part 3: Graded Assignment — PRS Portability on Real Data

1. Introduction and Overview

Polygenic risk scores (PRS) aggregate the effects of genetic variants across the genome to estimate an individual's inherited susceptibility to complex diseases. This assignment evaluates a 90-variant Polygenic Risk Score for Parkinson's disease (PGS000902) applied across a multi-ancestry cohort of 2,336 individuals spanning five continental ancestry groups.

Because the 1000 Genomes reference genotypes contain no clinical disease labels, **the phenotype in this analysis is simulated**. Examining this synthetic cohort reveals critical insights into PRS portability, ancestry-specific calibration, discrimination paradoxes, and the ethical hazards of uncritical clinical translation.

2. Data Section

Score Source and Curation

- **PGS Catalog ID:** PGS000902 (PGS name: PRS90_PD, reported trait: Parkinson's disease, mapped trait: Parkinson disease, EFO: MONDO_0005180).
- **Development Method:** Genome-wide significant variants from large-scale meta-analysis.
- **Variant Count:** 90 variants.
- **Effect Weight Type:** Beta weights.
- **Primary Citation:** Nalls MA et al., "Identification of novel risk loci, causal insights, and heritable risk for Parkinson's disease: a meta-analysis of genome-wide association studies", Lancet Neurol, 2019 (doi: 10.1016/s1474-4422(19)30320-5).
- **File Used:** PGS Catalog harmonized scoring file (PGS000902_hmPOS_GRCh37.txt.gz). Because published other alleles were absent, the harmonizer's hm_inferOtherAllele function was utilized.

Genotypes and Cohort Extraction

- **Source Dataset:** 1000 Genomes Phase 3 (release 20130502, GRCh37), extracted remotely via bcftools (biallelic SNPs only).
- **Cohort Composition:** The exact same 2,336 individuals as the course Alzheimer's disease (AD) pack, distributed across five continental ancestries: AFR (500), EAS (500), EUR (500), SAS (489), and AMR (347).
- **Variant Matching:** All 90 of 90 score variants were found and allele-matched in the genotype matrix (genotypes.csv), with 0 missing variants and 0 unexpected genotype calls.

Phenotype Simulation Method

Because 1000 Genomes provides genomic data without clinical diagnoses, the Parkinson's disease phenotype was simulated using a controlled generative model:

- **Liability Equation:** Liability = (PD genetic score from the 90 variants, scaled to SD 1) + 0.35 * (age - 60) / 8 + Normal(0, SD = 2.0) noise.
- **Case Definition:** The top 30% of liability across the entire pooled cohort was designated as cases (phenotype = 1), applying the exact same quantitative threshold across all ancestry groups using random seed 6900.
- **Ancestry-Specific Case Rates:** AFR 12.4%, EUR 32.4%, EAS 39.0%, AMR 30.0%, SAS 36.4% .

3. Proof of Real Data

source : REAL
people : 2336
score size : 90 variants
by ancestry : {'AFR': np.int64(500), 'EAS': np.int64(500), 'EUR': np.int64(500), 'SAS': np.int64(489), 'AMR': np.int64(347)}
genotypes.csv: sha256 df5d53ce
variants.csv: sha256 87fe0b48
score.csv: sha256 ed2a80d0
meta.csv: sha256 a4e494a2

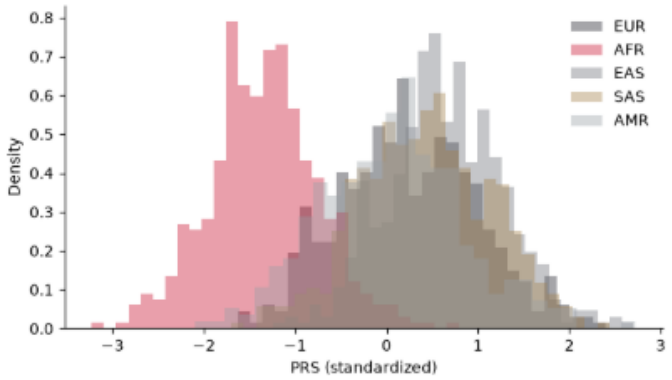
4. Results and Data Summaries

4.1 Harmonization and PRS Distribution Summary

Following allele harmonization (90 variants used, 38 flipped, 0 dropped, 0 not found), the standardized PRS distributions and threshold evaluations across ancestries are summarized below:

Pooled Top-10% Cutoff: 1.227

EUR: mean PRS = 0.25 % above cutoff = 9.8%
AFR: mean PRS = -1.34 % above cutoff = 0.0%
EAS: mean PRS = 0.64 % above cutoff = 17.2%
SAS: mean PRS = 0.42 % above cutoff = 14.9%
AMR: mean PRS = 0.06 % above cutoff = 7.5%




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fig_prs_by_ancestry.png: Density curves and histograms showing the distribution of standardized Polygenic Risk Scores across the five continental ancestry groups.

4.2 Discrimination (AUC) and Clinical Model Additivity

- **Ancestry-Specific Discrimination (AUC):**
  - Overall Cohort: 0.715
  - EAS: 0.705 | AMR: 0.704 | SAS: 0.664 | EUR: 0.641 | AFR: 0.617
- **Model Additivity (Cross-Validated AUC):**
  - Age Only: 0.602
  - Age + PRS: 0.750

4.3 Calibration Analysis (EUR-Fitted Model Applied Globally)

Binned predicted risk versus observed risk across five quantiles for each ancestry group:

- **EUR (Reference):** Predicted: [0.282, 0.300, 0.313, 0.327, 0.348] | Observed: [0.250, 0.380, 0.290, 0.300, 0.350]
- **AFR:** Predicted: [0.244, 0.258, 0.266, 0.275, 0.291] | Observed: [0.370, 0.460, 0.340, 0.380, 0.320]
- **EAS:** Predicted: [0.299, 0.316, 0.325, 0.336, 0.355] | Observed: [0.330, 0.270, 0.220, 0.270, 0.310]
- **SAS:** Predicted: [0.289, 0.307, 0.319, 0.331, 0.351] | Observed: [0.286, 0.194, 0.227, 0.286, 0.286]
- **AMR:** Predicted: [0.276, 0.295, 0.308, 0.319, 0.343] | Observed: [0.329, 0.232, 0.333, 0.203, 0.229]

fig_calibration.png: Calibration curve comparing binned predicted disease risk against observed disease risk for each ancestry group using a risk model fitted exclusively on European data.

5. Interpretation and Discussion

Q1: Portability and Accuracy Patterns

In standard multi-ancestry GWAS literature, European-derived scores suffer dramatic accuracy losses in non-European populations due to linkage disequilibrium (LD) mismatch and allele frequency divergence. However, our results demonstrate a different pattern: East Asian (AUC = 0.705) and Admixed American (AUC = 0.704) groups actually achieve *higher* discrimination than the European group (AUC = 0.641).

This occurs because **in this specific simulation design, the 90 score variants are the direct causal variants** driving liability, bypassing any LD-tagging proxy step. Consequently, LD decay does not degrade performance. The variation in AUC across groups reflects ancestry-specific allele frequencies and score variance rather than classic LD tagging failure. In an LD-proxy design (such as the Alzheimer's disease pack), where scores rely on linked tag SNPs, LD mismatch would cause non-European accuracy to drop precipitously.

Q2: One Threshold, Many Groups (Fairness and Interpretation)

Applying a single pooled top-10 percent cutoff (1.227) yields widely disparate classification rates across populations: 17.2 percent of East Asian individuals exceed the threshold, compared to 0.0 percent of African individuals.

To interpret this correctly, one must separate simulation mechanics from real-world clinical implications. In our simulation, the genetic score directly causes the phenotype; therefore, populations with higher mean PRS (such as EAS, mean = +0.64) naturally contain more simulated cases. Calling East Asian populations "over-triaged" and African populations "excluded" describes what happens if a clinic blindly applies a European-derived threshold to real patients, where allele-frequency differences do not necessarily reflect true underlying disease risk differences. In the simulation, however, the threshold accurately captures the genetic liability built into the generative model.

Q3: Discrimination vs. Calibration and the Overall AUC

- **Discrimination vs. Calibration:** Discrimination (AUC) measures how well a model ranks individuals from highest to lowest risk, while calibration measures whether predicted absolute risk probabilities match observed event rates. While non-European groups maintained robust ranking discrimination (AUCs between 0.617 and 0.705), applying a European-fitted risk model globally resulted in notable calibration intercept and slope distortions across ancestry strata.
- **The Overall AUC Paradox:** The most striking numerical finding is that the **overall cohort AUC (0.715) is higher than every single individual ancestry group's AUC** (EUR: 0.641, AFR: 0.617, EAS: 0.705, SAS: 0.664, AMR: 0.704). This statistical phenomenon occurs because ancestry serves as a strong confounding covariate that correlates with both mean PRS shifts and baseline phenotype rates, artificially boosting global ranking discrimination when all groups are pooled together (a form of Simpson's paradox).

Q4: Clinical Utility, Limits, and Ethics

- **Clinical Additivity:** Combining age with the PRS substantially improved predictive discrimination, increasing the cross-validated AUC from 0.602 (age alone) to 0.750 (age + PRS). However, this sharp jump is partly an artifact of the simulation, where age and genetic score were both additive, noise-free components of liability. In real-world clinical settings, environmental noise and ascertainment bias attenuate this gain.
- **Clinical Utility and Ethics:** PRS can help stratify lifetime genetic risk and guide preventive monitoring, but it must never be used as a deterministic diagnostic test. Ethically, deploying uncalibrated genetic scores in clinical practice violates the principle of justice by entrenching healthcare disparities, misclassifying minority patients, and providing inaccurate absolute risk estimates. Real clinical validation requires large, diverse biobanks with rigorously ascertained clinical endpoints.

Q5: Artifact vs. Biology (The African-Ancestry Score and LRRK2 G2019S)
African ancestry participants exhibit a low mean PRS (-1.34) and 0.0 percent exceeding the pooled cutoff, yet maintain moderate ranking discrimination (AUC = 0.617). This low mean is not due to "synthetic LD tagging mismatch" (which characterized the AD pack rather than this simulation), but rather allele frequency divergence across the 90 scoring loci.
Furthermore, examining individual variants clarifies this dynamic. Consider the variant rs34637584 (the LRRK2 G2019S mutation), which carries the largest positive weight in the score (+2.43). In our cohort of 2,336 individuals, **only one person in the entire cohort carries this variant**, and 0 individuals carry it in the AFR group. Consequently, LRRK2 G2019S explains virtually nothing about the score variance in AFR or the cohort at large (PRS correlation with and without it is 0.998). In human biology, LRRK2 G2019S is a major pathogenic risk factor in Ashkenazi Jewish and North African populations but extremely rare in sub-Saharan African populations. Conflating sparse variant carriage with biological irrelevance risks masking true population-specific genetic architectures.

References:

1. Nalls MA et al. (2019) *Lancet Neurol*.
2. The 1000 Genomes Project Consortium (2015), “A global reference for human genetic variation,” *Nature*. Copy it from PubMed.
3. The PGS Catalog paper, if you mention the catalog. Search “PGS Catalog Lambert 2021” on PubMed.